EuroSkills 2027 – IT Network Systems Administration

National Semi-final

Module C

28th February 2026

About the project

The German brewery, **Paulaner** was quite satisfied with your previous work, so they have asked you again to complete a project for them. This time, they are planning on deploying a georedundant data center to host their applications. Your task will be implementing a proof-of-concept design of this network, including just one data center site. The application data stored on the server is encrypted, and the decryption keys are stored in hardware security modules on an offsite server, to be compliant with data protection laws. ***Viel Glück!***

Preliminary Notice

Please read the instructions carefully to make sure you receive the highest mark. We advise you to read all the instructions before beginning to work.

The ISP device has already been configured, and you do not have access to it. The endpoint devices are also preconfigured; the credentials are **cisco/cisco**.

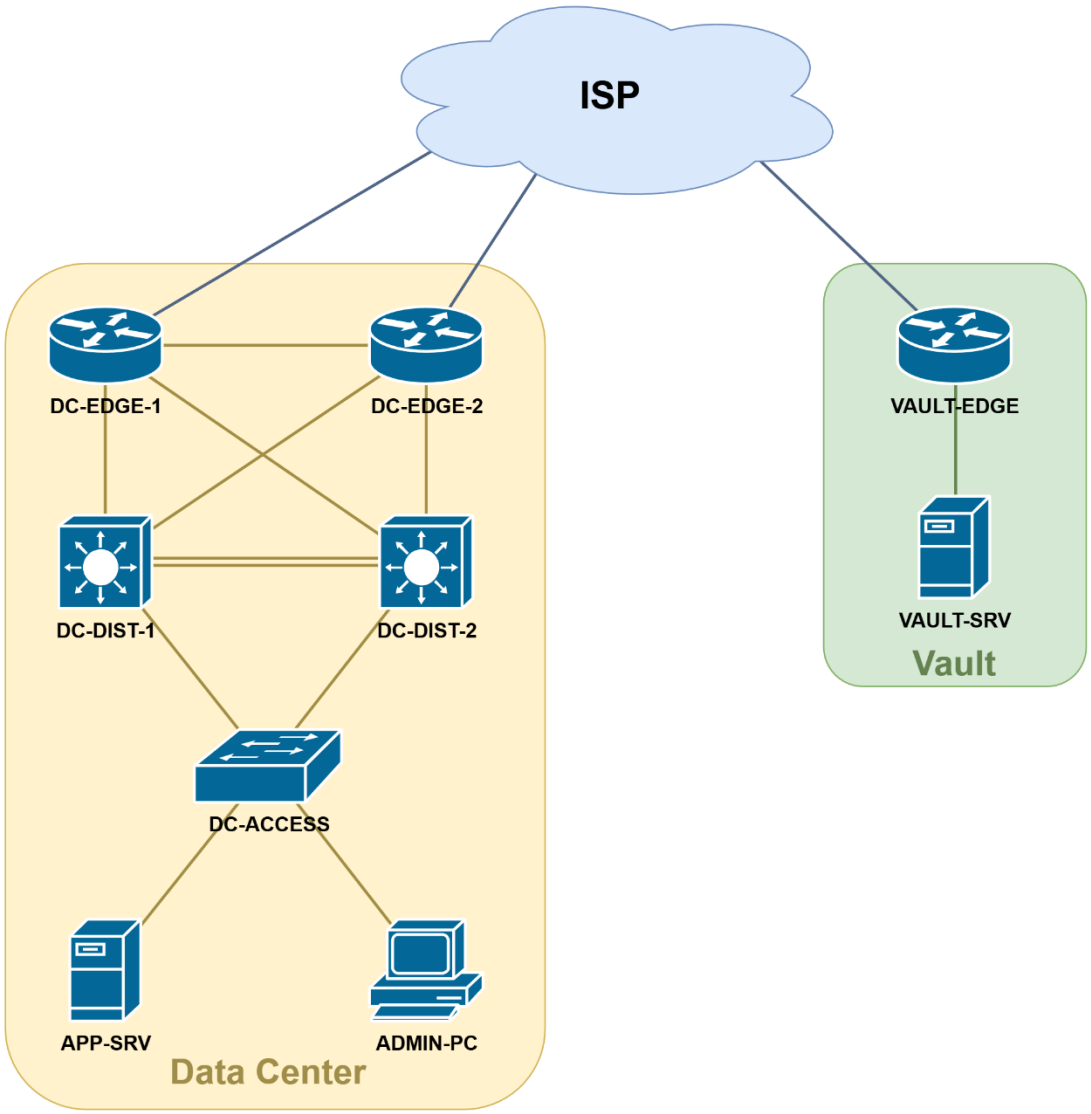
On completion of the test project, please leave all devices in the enabled exec menu!

Whenever a parameter *(e.g. EtherChannel number)* is not specified, feel free to choose your own, as the choice you make will not affect the marking. However, when it is specified, make sure to use it!

Please note that the marking is functional, thus we will not check your configuration, only the functionality of your network.

You can access the CML instance at **https://cml.s39.lan**

Topology


ISP
DC-EDGE-1
DC-EDGE-2
DC-DIST-1
DC-DIST-2
DC-ACCESS
APP-SRV
ADMIN-PC
Data Center
VAULT-EDGE
VAULT-SRV
Vault

IP Addressing

Please configure device interfaces according to the table below.

Device	Interface	IP Address
DC-EDGE-1	G0/0	10.1.0.1/30
	G0/1	10.1.0.5/30
	G0/2	10.1.0.17/30
	G0/3	93.192.96.49/30
	Loopback0	10.1.255.1/32
DC-EDGE-2	G0/0	10.1.0.9/30
	G0/1	10.1.0.13/30
	G0/2	10.1.0.18/30
	G0/3	93.192.96.53/30
	Loopback0	10.1.255.2/32
DC-DIST-1	G0/0	10.1.0.2/30
	G0/1	10.1.0.10/30
	Vlan100	159.60.131.188/28
	Vlan600	10.1.6.252/24
DC-DIST-2	G0/0	10.1.0.6/30
	G0/1	10.1.0.14/30
	Vlan100	159.60.131.189/28
	Vlan600	10.1.6.253/24
VAULT-EDGE	G0/1	10.0.0.254/24
	G0/3	93.192.96.193/30

Furthermore, endpoint devices are already configured with the following addresses:

Device	IP Address
APP-SRV	159.60.131.184/28
ADMIN-PC	10.1.6.1/24
VAULT-SRV	10.0.0.1/24

Data Center Network

Access networks

Configure a layer-2 EtherChannel between **DC-DIST-1** and **DC-DIST-2**. Use a vendor-agnostic protocol for negotiation.

Configure the following VLANs on all the switches:

VLAN Name	VLAN ID
APPLICATION	100
ADMIN	600
NATIVE	999

Set up trunk interfaces between the switches. The trunks should only allow traffic of **APPLICATION** and **ADMIN** VLANs. Use VLAN **999** for native VLAN, but make sure the frames in this VLAN are tagged, too.

On **DC-ACCESS**, set up access interfaces for the endpoints, according to their names. Configure the following technologies on the access interfaces:

- BPDU Guard
- PortFast
- Port security
    - Maximum **1** address, learned & saved in the configuration
    - When a violation occurs, do not shut down the interface, but block traffic, and generate a syslog message

Configure **Rapid Spanning Tree Protocol** on the L2 network. By default, **DC-DIST-1** should be the root bridge, and in case of failure, **DC-DIST-2** should take over.

Configure **HSRP** to provide redundant gateways for the endpoint networks. Make sure **DC-DIST-1** has a higher priority than **DC-DIST-2**. After a failure is resolved, make sure **DC-DIST-1** regains the active role. Use the highest unicast IP address for the virtual IP.

Transit networks

Configure **EIGRP** routing between the routers and the distribution layer switches. Use **AS number 1**. Advertise **all** loopback addresses and networks, except the outside *(G0/3)* interfaces of the edge routers. Do not leak routing information into endpoint networks.

Internet connectivity

Configure **BGP** sessions with the ISP on **all three edge routers**. The ISP, ***Deutsche Telekom AG*** is putting efforts into making their transit network more robust, so they now implement BFD for faster convergence. Use **BFD** to detect peer failure. Refer to the following table for more information.

Device name	Device AS	Peer address	Peer AS	Session password	BFD timers
DC-EDGE-1	35281	93.192.96.50	3320	TelekomAmBesten	**Tx:** 50ms **Min_rx:** 50ms **Multiplier:** 3
DC-EDGE-2		93.192.96.54			
VAULT-EDGE	35280	93.192.96.194			

The ISP will provide a default route prefix, along with a lot of prefixes which you do not need. Filter the incoming prefixes to only allow the default route.

From both **DC-EDGE** routers, advertise the application network ***(159.60.131.176/28)*** in BGP. Make sure that the ISP does not receive any private *(RFC1918)* addresses. From BGP, redistribute the **default route into the EIGRP** process.

Set up an **iBGP** peering between **DC-EDGE-1** and **DC-EDGE-2**. Use their **Loopback0** addresses for peering. **Set next-hop-self on the neighborships.**

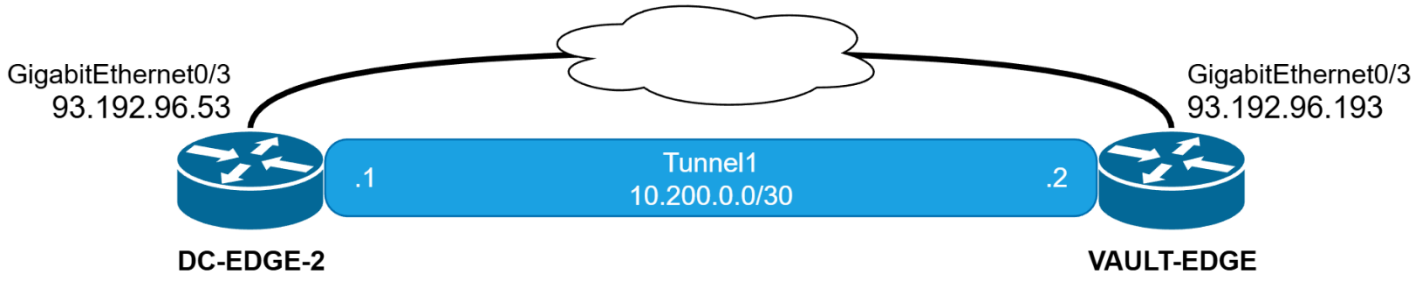
DC-EDGE-1		DC-EDGE-2	
Source interface	**IP Address**	**IP Address**	**Source interface**
Loopback0	10.1.255.1	10.1.255.2	Loopback0

Secure tunnel

To facilitate connectivity between **APP-SRV** and **VAULT-SRV**, a secure tunnel will be needed between the sites. Configure a **GRE over IPSec** tunnel with **IKEv1** authentication between **DC-EDGE-2** and **VAULT-EDGE**. Refer to the following table for implementation details.

Phase 1 (Main mode)	Encryption	AES-256
	Integrity algorithm	SHA-512
	Key exchange	Diffie-Hellman group 21
	SA Lifetime	3600 seconds
	Pre-shared key	PAULANER
Phase 2 (Quick mode)	Encryption	AES-256
	Integrity algorithm	HMAC-SHA512
	SA Lifetime	3600 seconds

Use the figure below for the tunnel configuration.



Extend EIGRP routing through the tunnel so the two sites can reach each other.

Locking down vault network

VAULT-SRV uses *Key Management Interoperability Protocol (KMIP)* for exchanging encryption keys. The service is listening on port **TCP/5696**.

Apply ACLs in both directions of the **VAULT-EDGE G0/1** interface. The only traffic that should be permitted for the whole Vault network is:

- **APP-SRV** reaching the *KMIP* service on **VAULT-SRV**
- Return traffic (stateful) for **APP-SRV** from the *KMIP* service – only for connections already initiated by **APP-SRV**

IP Addresses of the devices are:

- **APP-SRV:** 159.60.131.184
- **VAULT-SRV:** 10.0.0.1

Testing

You can use the following test to make sure everything works correctly:

- **APP-SRV** can reach the ISP *(ping 8.8.8.8)*
- **APP-SRV** can reach the *KMIP* service on **VAULT-SRV**
  - To test this, enter the following command on **VAULT-SRV**:
    `nc -l -p 5696`
  - Then, on **APP-SRV**:
    `telnet 10.0.0.1 5696`
  - Now whatever you send from **APP-SRV** should appear on **VAULT-SRV**